

# BACnet Protocol Implementation Conformance Statement (PICS)

## Chipkin BACnet/SC Hub (B-SCHUB) example, version 1.4.0

Date: 2026-09-26. Source, manual and releases: <https://github.com/chipkin/BACnetProfileExample-B-SCHUB-CPP>.

The vendor, model and version rows below come from the CHANGE ALL OF THIS BEFORE YOU SHIP block at the top of main.cpp. If you build your own product from this example, update that block and this document. This device has **not** been submitted for BTL certification.

## 1. Product description

|                                     |                                         |
|-------------------------------------|-----------------------------------------|
| <b>Vendor Name</b>                  | Chipkin Automation Systems              |
| <b>Vendor Identifier</b>            | 389                                     |
| <b>Product Name</b>                 | CAS BACnet Stack Example - B-SCHUB      |
| <b>Product Model Number</b>         | CAS BACnet Stack Example - B-SCHUB      |
| <b>Application Software Version</b> | 1.4.0                                   |
| <b>Firmware Revision</b>            | 6.0.23.0 (the CAS BACnet Stack version) |
| <b>BACnet Protocol Version</b>      | 1                                       |
| <b>BACnet Protocol Revision</b>     | 30                                      |

**Product Description:** a BACnet Secure Connect hub. BACnet/SC devices connect to it over TLS 1.3 WebSockets with mutual certificate authentication, and it relays their traffic. It can also connect out to another hub as a node. A BACnet/IP port stays active for discovery and management. Its certificates can be replaced over BACnet with the BACnet/SC certificate procedures. Built on the CAS BACnet Stack.

## 2. BACnet standardized device profile (Annex L)

### B-SCHUB - BACnet Secure Connect Hub.

This device claims exactly one profile. Because the B-SCHUB requirements are a superset of B-GENERAL's, a conformant B-SCHUB device also satisfies **B-GENERAL** (Annex L.8); that is subsumption, not a second claim.

## 3. BIBBs supported (Annex K)

| BIBB     | Description                                        | Notes                  |
|----------|----------------------------------------------------|------------------------|
| DS-RP-B  | Data Sharing - ReadProperty - B                    |                        |
| DS-RPM-B | Data Sharing - ReadPropertyMultiple - B            |                        |
| DM-DDB-B | Device Management - Dynamic Device Binding - B     |                        |
| DM-DOB-B | Device Management - Dynamic Object Binding - B     |                        |
| DM-DCC-B | Device Management - DeviceCommunicationControl - B | optional password      |
| NM-SCH-B | Network Management - BACnet/SC Hub Function - B    | Network Port 2; see §9 |

No other BIBBs are supported. In particular this device does **not** claim DS-WP-B, DS-COV-B, any alarm and event (AE-) *BIBB*, *scheduling* (SCHED-), or trending (T-\*).

It executes the device-B side of the BACnet/SC certificate procedures (135-2024 clause 19.8.3): WriteProperty File\_Size and AtomicWriteFile into the operational and issuer certificate File objects, applied by ReinitializeDevice ACTIVATE\_CHANGES/WARMSTART after validation, and the Network Port Command values DISCARD\_CHANGES (drops staged certificate writes) and GENERATE\_CSR\_FILE (new key pair and certificate signing request, Network Port 2).

## 4. Application services supported

| Service                    | Initiate   | Execute                                                                        |
|----------------------------|------------|--------------------------------------------------------------------------------|
| ReadProperty               | no         | <b>yes</b>                                                                     |
| ReadPropertyMultiple       | no         | <b>yes</b>                                                                     |
| Who-Is                     | no         | <b>yes</b>                                                                     |
| I-Am                       | <b>yes</b> | -                                                                              |
| Who-Has                    | no         | <b>yes</b>                                                                     |
| I-Have                     | <b>yes</b> | -                                                                              |
| DeviceCommunicationControl | no         | <b>yes</b>                                                                     |
| AtomicReadFile             | no         | <b>yes</b>                                                                     |
| AtomicWriteFile            | no         | <b>yes</b> (certificate File objects 1, 3, 4)                                  |
| WriteProperty              | no         | <b>yes</b> (File_Size of File objects 1, 3, 4, and Network Port Command, only) |
| ReinitializeDevice         | no         | <b>yes</b> (ACTIVATE_CHANGES, WARMSTART only)                                  |

An unsolicited I-Am is broadcast at start-up, as well as in response to Who-Is.

Any other confirmed service is rejected, and WriteProperty to any other property is refused with write-access-denied.

## 5. Segmentation capability

Segmentation is **not supported** in either direction (Segmentation\_Supported = no-segmentation). Max\_APDU\_Length\_Accepted is 1476 octets, the BACnet/IP maximum.

## 6. Standard object types supported

No object is dynamically creatable or deletable. The only writable properties are File\_Size of File objects 1, 3 and 4 and Command of both Network Port objects (certificate procedures).

| Object type  | Instance | Object_Name                 | Optional properties supported |
|--------------|----------|-----------------------------|-------------------------------|
| Device       | 389022   | Chipkin Example B-SCHUB     | Description                   |
| Analog Input | 1        | Bronze                      | Description                   |
| Network Port | 1        | BACnet IP                   | Description                   |
| Network Port | 2        | BACnet SC                   | Description                   |
| File         | 1        | Operational Certificate     | Description                   |
| File         | 2        | Certificate Signing Request | Description                   |
| File         | 3        | Issuer Certificate Slot 1   | Description                   |
| File         | 4        | Issuer Certificate Slot 2   | Description                   |

The device instance is configurable with --deviceID or the config file's device-id (BACnet requires the device instance to be configurable).

## 7. Data link layer options

**BACnet/IP (Annex J)**, UDP port 47808 (0xBAC0) by default, configurable with --port. Active by default; --bacnet-ip off (config bacnet-ip = off) removes it, leaving a BACnet/SC-only device without Network Port 1.

**BACnet/SC (Annex AB)** on Network Port 2: hub function on wss://0.0.0.0:47819/ by default (configurable with --sc-port), and an optional hub connector (--sc-hub-uri, --sc-failover-uri). TLS 1.3 with mutual authentication, WebSocket subprotocol hub.bsc.bacnet.org.

BBMD is not supported, Foreign Device registration is not supported, and MS/TP, Ethernet (Annex H) and PTP are not supported.

## 8. Device address binding

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Static device binding is **not supported**. The device answers Who-Is/Who-Has and does not initiate any confirmed request, so it never needs to bind a peer.

## 9. Networking options

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Not a router, not a BBMD, and does not register as a foreign device. Each Network Port reports Network\_Number 0 with Network\_Number\_Quality unknown, unless --ip-network-number / --sc-network-number (config ip-network-number / sc-network-number) sets it; it is then reported with quality configured.

The Device's Object\_Name is configurable with --device-name (config device-name), so each installed hub can have a unique name.

**BACnet/SC hub function (NM-SCH-B):** accepts at most 4 connections at once (a fixed limit of this example; --sc-max-hub-connections can lower it) and relays unicast and broadcast BVLC-SC messages between them and the local device. Connecting devices must present a certificate that chains to an issuer in File 3 or File 4. Certificate revocation lists are not checked. The hub connector verifies the remote hub's certificate chain but not its host name (BACnet/SC certificates identify devices, not DNS names).

## 10. Character sets supported

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UTF-8 (ANSI X3.4). Supporting a character set does not imply the device can handle data in all character sets.

## 11. Objects and properties

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Every object this example creates, and every REQUIRED property of each (per ANSI/ASHRAE 135-2024 clause 12 and the stack's docs/property-profile-reference.md), plus the optional properties the example turns on. **Served by** says who answers a ReadProperty: the **stack** generates it, or the **app** serves it from a GetProperty\* callback in main.cpp. A  $\Delta$  row is a required property the app does not serve and the stack would fill with a default - that is a defect, not a feature.

**Device 389022 "Chipkin Example B-SCHUB" - vendor 389 (Chipkin Automation Systems); instance configurable with --deviceId and name with --device-name. The stack rows are device-wide facts only the stack knows - the protocol version and revision it implements, the services and object types it was configured with, the live object list and address-binding table. The accepted rows are the stack's configured defaults for APDU limits, segmentation, system status and database revision; an application that answered them from its own constants could contradict the stack, so this example does not**

| Property                                 | Datatype                   | Served by                                               | Writable |
|------------------------------------------|----------------------------|---------------------------------------------------------|----------|
| Object_Identifier                        | BACnetObjectIdentifier     | stack                                                   | no       |
| Object_Name                              | CharacterString            | app                                                     | no       |
| Object_Type                              | BACnetObjectType           | stack                                                   | no       |
| System_Status                            | BACnetDeviceStatus         | stack default, accepted (Generic Enumerated default: 0) | no       |
| Vendor_Name                              | CharacterString            | app                                                     | no       |
| Vendor_Identifier                        | Unsigned16                 | app                                                     | no       |
| Model_Name                               | CharacterString            | app                                                     | no       |
| Firmware_Revision                        | CharacterString            | app                                                     | no       |
| Application_Software_Version             | CharacterString            | app                                                     | no       |
| Description ( <i>optional, enabled</i> ) | CharacterString            | app                                                     | no       |
| Protocol_Version                         | Unsigned                   | stack                                                   | no       |
| Protocol_Revision                        | Unsigned                   | stack                                                   | no       |
| Protocol_Services_Supported              | BACnetServicesSupported    | stack                                                   | no       |
| Protocol_Object_Types_Supported          | BACnetObjectTypesSupported | stack                                                   | no       |

| Property                 | Datatype                                   | Served by                                                                       | Writable |
|--------------------------|--------------------------------------------|---------------------------------------------------------------------------------|----------|
| Object_List              | BACnetARRAY[N] of BACnetObjectIdentifier   | stack                                                                           | no       |
| Max_APDU_Length_Accepted | Unsigned                                   | stack default, accepted<br>(CAS_BACNET_DEVICE_DEFAULT_MAX_APDU_LENGTH_ACCEPTED) | no       |
| Segmentation_Supported   | BACnetSegmentation                         | stack default, accepted<br>(BACnetSegmentation::noSegmentation)                 | no       |
| APDU_Timeout             | Unsigned                                   | stack default, accepted<br>(CAS_BACNET_DEVICE_DEFAULT_APDU_TIMEOUT)             | no       |
| Number_Of_APDU_Retries   | Unsigned                                   | stack default, accepted<br>(CAS_BACNET_DEVICE_DEFAULT_NUMBER_OF_APDU_RETRIES)   | no       |
| Device_Address_Binding   | BACnetLIST of BACnetAddressBinding         | stack                                                                           | no       |
| Database_Revision        | Unsigned                                   | stack default, accepted (Generic UnsignedInteger default: 0)                    | no       |
| Property_List            | BACnetARRAY[N] of BACnetPropertyIdentifier | stack                                                                           | no       |

**Analog Input 1 "Bronze" - REAL, degrees Celsius; starts at 21.5; read-only (example data, changed with the arrow keys)**

| Property                                 | Datatype                                   | Served by                                               | Writable |
|------------------------------------------|--------------------------------------------|---------------------------------------------------------|----------|
| Object_Identifier                        | BACnetObjectIdentifier                     | stack                                                   | no       |
| Object_Name                              | CharacterString                            | app                                                     | no       |
| Object_Type                              | BACnetObjectType                           | stack                                                   | no       |
| Present_Value                            | Real                                       | app                                                     | no       |
| Description ( <i>optional, enabled</i> ) | CharacterString                            | app                                                     | no       |
| Status_Flags                             | BACnetStatusFlags                          | stack                                                   | no       |
| Event_State                              | BACnetEventState                           | stack default, accepted (Generic Enumerated default: 0) | no       |
| Out_Of_Service                           | Boolean                                    | app                                                     | no       |
| Units                                    | BACnetEngineeringUnits                     | app                                                     | no       |
| Property_List                            | BACnetARRAY[N] of BACnetPropertyIdentifier | stack                                                   | no       |

**Network Port 1 "BACnet IP" - BACnet/IP; kept active throughout so this example stays discoverable over plain BACnet/IP regardless of the BACnet/SC transport outcome below, unless --bacnet-ip off (BACnet/SC only) leaves this object out altogether. Network\_Type and Protocol\_Level are set from BACnetStack\_AddNetworkPortObject()'s arguments (IPv4, BACnet Application) at start-up, not a GetProperty callback like the object's other app-served rows; Changes\_Pending is likewise computed and answered natively by the stack's Network Port object. Reliability has no fault condition this example detects, so it is accepted at the generic default (normal). Network\_Number is 0 with Network\_Number\_Quality unknown unless --ip-network-number sets it (then quality configured); the device is not a router. Command (cl. 12.56.16) is present and writable, executed by the stack: DISCARD\_CHANGES reverts a pending Network\_Number**

| Property          | Datatype               | Served by | Writable |
|-------------------|------------------------|-----------|----------|
| Object_Identifier | BACnetObjectIdentifier | stack     | no       |
| Object_Name       | CharacterString        | app       | no       |
| Object_Type       | BACnetObjectType       | stack     | no       |

| Property                                 | Datatype                                   | Served by                                               | Writable |
|------------------------------------------|--------------------------------------------|---------------------------------------------------------|----------|
| Description ( <i>optional, enabled</i> ) | CharacterString                            | app                                                     | no       |
| Status_Flags                             | BACnetStatusFlags                          | stack                                                   | no       |
| Reliability                              | BACnetReliability                          | stack default, accepted (Generic Enumerated default: 0) | no       |
| Out_Of_Service                           | Boolean                                    | app                                                     | no       |
| Network_Type                             | BACnetNetworkType                          | app                                                     | no       |
| Protocol_Level                           | BACnetProtocolLevel                        | app                                                     | no       |
| Changes_Pending                          | Boolean                                    | app                                                     | no       |
| Command                                  | BACnetNetworkPortCommand                   | stack                                                   | yes      |
| Property_List                            | BACnetARRAY[N] of BACnetPropertyIdentifier | stack                                                   | no       |

**Network Port 2 "BACnet SC" - BACnet/SC (Network\_Type = secureConnect(11)); hosts the NM-SCH-B hub function (listener) and the optional hub connector; the WebSocket/TLS transport is sc\_transport/ScTransport.**

**Network\_Type/Protocol\_Level are set from BACnetStack\_AddNetworkPortObject()'s arguments at start-up, same as Network Port 1. Reliability, Network\_Number and Network\_Number\_Quality as for Network Port 1. Command (cl. 12.56.16) is writable: DISCARD\_CHANGES also drops staged certificate writes, and GENERATE\_CSR\_FILE makes a new key pair and Certificate Signing Request (File 2) - the NetworkPortCommand callback in main.cpp**

| Property                                 | Datatype                                   | Served by                                               | Writable |
|------------------------------------------|--------------------------------------------|---------------------------------------------------------|----------|
| Object_Identifier                        | BACnetObjectIdentifier                     | stack                                                   | no       |
| Object_Name                              | CharacterString                            | app                                                     | no       |
| Object_Type                              | BACnetObjectType                           | stack                                                   | no       |
| Description ( <i>optional, enabled</i> ) | CharacterString                            | app                                                     | no       |
| Status_Flags                             | BACnetStatusFlags                          | stack                                                   | no       |
| Reliability                              | BACnetReliability                          | stack default, accepted (Generic Enumerated default: 0) | no       |
| Out_Of_Service                           | Boolean                                    | app                                                     | no       |
| Network_Type                             | BACnetNetworkType                          | app                                                     | no       |
| Protocol_Level                           | BACnetProtocolLevel                        | app                                                     | no       |
| Changes_Pending                          | Boolean                                    | app                                                     | no       |
| Command                                  | BACnetNetworkPortCommand                   | stack                                                   | yes      |
| Property_List                            | BACnetARRAY[N] of BACnetPropertyIdentifier | stack                                                   | no       |

**File 1 "Operational Certificate" - writable over BACnet (clause 19.8.3): File\_Size and AtomicWriteFile, staged until ReinitializeDevice ACTIVATE\_CHANGES/WARMSTART, which validates the set (parses, matches the hub's private key, chains to an issuer) before writing operational-certificate.pem and reloading TLS. Serves the hub's operational certificate via AtomicReadFile (stream access) - bound to Network Port 2's Operational\_Certificate\_File. File\_Size/Modification\_Date come from the staged copy or the file on disk**

| Property          | Datatype               | Served by | Writable |
|-------------------|------------------------|-----------|----------|
| Object_Identifier | BACnetObjectIdentifier | stack     | no       |
| Object_Name       | CharacterString        | app       | no       |
| Object_Type       | BACnetObjectType       | stack     | no       |

| Property                                 | Datatype                                   | Served by | Writable |
|------------------------------------------|--------------------------------------------|-----------|----------|
| Description ( <i>optional, enabled</i> ) | CharacterString                            | app       | no       |
| File_Type                                | CharacterString                            | app       | no       |
| File_Size                                | Unsigned                                   | app       | yes      |
| Modification_Date                        | BACnetDateTime                             | app       | no       |
| Archive                                  | Boolean                                    | app       | no       |
| Read_Only                                | Boolean                                    | app       | no       |
| File_Access_Method                       | BACnetFileAccessMethod                     | stack     | no       |
| Property_List                            | BACnetARRAY[N] of BACnetPropertyIdentifier | stack     | no       |

**File 2 "Certificate Signing Request" - read-only; serves the hub's certificate signing request (certificate-signing-request.pem) - bound to Network Port 2's Certificate\_Signing\_Request\_File. rewritten by the hub on Network Port 2 Command GENERATE\_CSR\_FILE, for a new key that replaces the hub's key when a certificate signed for it is activated**

| Property                                 | Datatype                                   | Served by | Writable |
|------------------------------------------|--------------------------------------------|-----------|----------|
| Object_Identifier                        | BACnetObjectIdentifier                     | stack     | no       |
| Object_Name                              | CharacterString                            | app       | no       |
| Object_Type                              | BACnetObjectType                           | stack     | no       |
| Description ( <i>optional, enabled</i> ) | CharacterString                            | app       | no       |
| File_Type                                | CharacterString                            | app       | no       |
| File_Size                                | Unsigned                                   | app       | no       |
| Modification_Date                        | BACnetDateTime                             | app       | no       |
| Archive                                  | Boolean                                    | app       | no       |
| Read_Only                                | Boolean                                    | app       | no       |
| File_Access_Method                       | BACnetFileAccessMethod                     | stack     | no       |
| Property_List                            | BACnetARRAY[N] of BACnetPropertyIdentifier | stack     | no       |

**File 3 "Issuer Certificate Slot 1" - writable over BACnet (clause 19.8.3), same staging as File 1; issuer certificate slot 1 (issuer-certificate.pem) - one of Network Port 2's 2 Issuer\_Certificate\_Files entries**

| Property                                 | Datatype                                   | Served by | Writable |
|------------------------------------------|--------------------------------------------|-----------|----------|
| Object_Identifier                        | BACnetObjectIdentifier                     | stack     | no       |
| Object_Name                              | CharacterString                            | app       | no       |
| Object_Type                              | BACnetObjectType                           | stack     | no       |
| Description ( <i>optional, enabled</i> ) | CharacterString                            | app       | no       |
| File_Type                                | CharacterString                            | app       | no       |
| File_Size                                | Unsigned                                   | app       | yes      |
| Modification_Date                        | BACnetDateTime                             | app       | no       |
| Archive                                  | Boolean                                    | app       | no       |
| Read_Only                                | Boolean                                    | app       | no       |
| File_Access_Method                       | BACnetFileAccessMethod                     | stack     | no       |
| Property_List                            | BACnetARRAY[N] of BACnetPropertyIdentifier | stack     | no       |

**File 4 "Issuer Certificate Slot 2" - writable over BACnet (clause 19.8.3), same staging as File 1; issuer certificate slot 2 (issuer-certificate-2.pem once written; until then it serves slot 1's certificate). TLS trusts every issuer in both slots (trusted-issuers.pem)**

| Property                                 | Datatype                                   | Served by | Writable |
|------------------------------------------|--------------------------------------------|-----------|----------|
| Object_Identifier                        | BACnetObjectIdentifier                     | stack     | no       |
| Object_Name                              | CharacterString                            | app       | no       |
| Object_Type                              | BACnetObjectType                           | stack     | no       |
| Description ( <i>optional, enabled</i> ) | CharacterString                            | app       | no       |
| File_Type                                | CharacterString                            | app       | no       |
| File_Size                                | Unsigned                                   | app       | yes      |
| Modification_Date                        | BACnetDateTime                             | app       | no       |
| Archive                                  | Boolean                                    | app       | no       |
| Read_Only                                | Boolean                                    | app       | no       |
| File_Access_Method                       | BACnetFileAccessMethod                     | stack     | no       |
| Property_List                            | BACnetARRAY[N] of BACnetPropertyIdentifier | stack     | no       |

## 12. References

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- ANSI/ASHRAE Standard 135-2024, Annex A (PICS template), Annex K (BIBBs), Annex L (device profiles), Annex AB (BACnet/SC), Clause 12 (object types).
- [README.md](#) - the manual.
- [TUTORIAL.md](#) - how to extend it, and how to keep this document honest when you do.